

Advanced (Habanero)

- Q1.** Evaluate the expression $2 \cdot 3 + 12 \div 4$ using the correct order of operations.
(A) 9
(B) 10
(C) 11
(D) 12
(E) 13
- Q2.** Which algebraic expression represents the phrase '5 more than 3 times a number' ?
(A) $3x + 5$
(B) $3x - 5$
(C) $5x + 3$
(D) $5x - 3$
(E) $x + 15$
- Q3.** Simplify the expression $2x + 5x$.
(A) $7x$
(B) $6x$
(C) $4x$
(D) $3x$
(E) x
- Q4.** Evaluate the expression $x^2 + 4x + 4$ when $x = 2$.
(A) 12
(B) 16
(C) 20
(D) 24
(E) 28
- Q5.** Which of the following expressions is equivalent to $9 - 3(2x - 1)$?
(A) $9 - 6x + 3$
(B) $9 - 6x - 3$
(C) $9 + 6x - 3$
(D) $9 + 6x + 3$
(E) $3 - 6x - 1$
- Q6.** Evaluate the expression $(2 + 3) \cdot (4 - 1)$.
(A) 9
(B) 10
(C) 11
(D) 12
(E) 15
- Q7.** Simplify the expression $3(2x - 1) + 2(x + 4)$.
(A) $8x + 5$
(B) $8x - 5$
(C) $6x + 5$
(D) $6x - 5$
(E) $4x + 3$
- Q8.** Evaluate the expression $12 \div (3 + 1) + 2$.
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $2x^2 + 5x - 3 + 3x^2 - 2x - 1$ and evaluate it when $x = -1$.

- Q10.** Translate the phrase 'the product of 2 more than a number and 3 less than the number' into an algebraic expression, then simplify and evaluate when the number is 5.

Chef's Tips

- Remind students to follow the order of operations (PEMDAS) when evaluating expressions.
- Encourage students to use algebraic properties to simplify expressions.
- Emphasize the importance of substituting values correctly into expressions.